Electric charges need a conductive path to flow. An electric circuit is a closed path to allow the flow of electrons. It usually includes a voltage source and a load. There can be other components as well to control the parameters of the circuit.

2.1 Basic Components of Electric Circuits

There are following basic components of electric circuits:

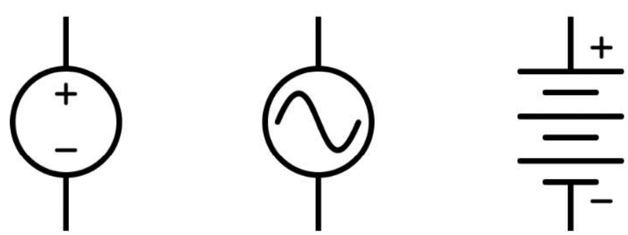
- Conducting Wire
- Voltage Source
- Load
- Resistors
- Capacitors

Conducting Wire

A conducting wire forms a path for the flow of electrons. It is used to connect different electrical components together. It should be a good conductor. Lesser the resistance of the wire, higher will be the conductance. It is a good practice to incorporate the resistance of conducting wire while solving an electric circuit.

Voltage Source

A voltage source is an essential component in an electric circuit. It acts as a source of electrons and hence causes current. There can be different types of voltage sources in a circuit i.e., battery, power supply and DC adapter etc. These sources can be AC or DC voltage sources. Different symbols used for the voltage source are given below:



Symbols of Voltage sources

Load

Load is the consumer of electric current. It can be an appliance i.e., light bulb and fan etc. Load can be resistive, capacitive and inductive. Current consumed in a circuit is decided by the load. Lower the resistance of the load, more will be the current.

Do you know?

All appliances in a home is always connected in parallel.

Point to Ponder

- Which types of load an energy saver bulb and electric fan are?

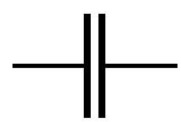
Resistors

Resistor is one of the basic elements of an electric circuit. They are commonly used to reduce the current flowing in a certain branch of an electric circuit. Its symbol is:

Symbol of Resistor

Capacitors

A capacitor is an electrical component to store electrical energy. It is created using two conductors placed next to each other and there is an insulator in between them. It stores electrical energy in its electric field. It blocks the DC but allows AC to pass through it. Its symbol is:



Symbol of Capacitor

2.1.1 Series Connections in Electric Circuits

In series connection, there is a single path for the current flow. All the components are connected end to end such that current flowing in the circuit remains the same. The sum of voltage across each element in a series circuit is equal to the total voltage of the power supply. Total resistance of the circuit is equal to the sum of individual resistances.

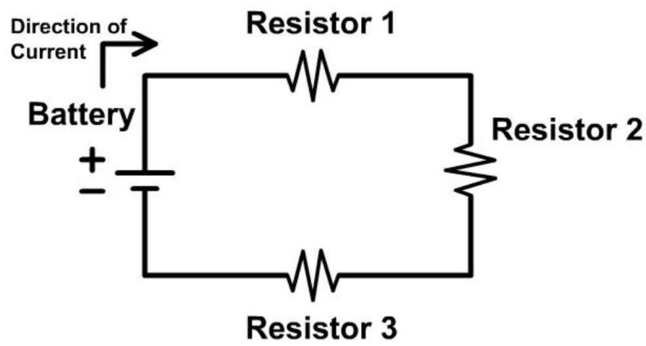


Fig. 2.1 Series Circuit

2.1.2 Parallel Connections in Electric Circuits

In a parallel connection, there can be multiple paths for the current flow. All the components are connected to a set of common points of the circuit such that the voltage across each component remains the same. The sum of current in each branch of a parallel circuit is equal to the total current flowing in the circuit. Total resistance of the circuit is less than the resistance of individual resistances.

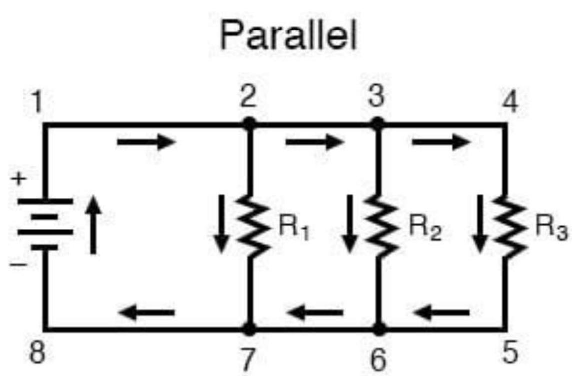


Fig. 2.2 Parallel Circuit

2.1.3 Series-Parallel Connections in Electric Circuits

When the components are connected in such a way that neither series rule nor parallel rule apply, then we have to identify individual portions of the circuit that are either connected in series or in parallel. Take the following circuit, for instance:

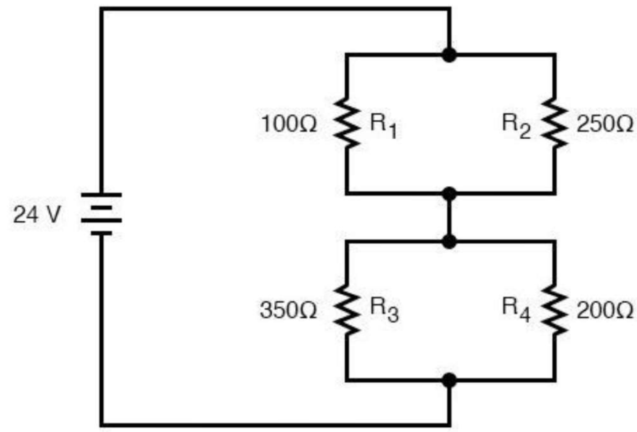


Fig. 2.3 Series Parallel Circuit

Here, R_1 and R_2 are connected in parallel. Similar is the case with R_3 and R_4. These two sets of resistance are then connected in series.

2.2 Kirchhoff's Voltage Law

Kirchhoff's Voltage Law (KVL) states that the sum of voltages inside a closed loop in an electric circuit is always equal to zero. In other words, the sum of all voltages across components which supply electrical energy (such as cells or generators) must equal the sum of all voltages across the other components (resistance, capacitors etc.) in the same loop. This law is a consequence of both charge conservation and the conservation of energy.

As charge carriers flowing through a circuit pass through a component, they either gain or lose electrical energy, depending upon the component (cell or resistor, for example). This is due to the fact that work is done on or by them as a result of the electric forces inside the components. The total work done on a charge carrier by electric forces in supply components (such as cells) must be equal to the total work done by the charge carrier in other components (such as resistors and lamps). This means that the sum of all potential differences across the components involved in a circuit's loop must be zero if we count voltages across supply components as positive and across 'electricity using' components as negative.

Figure 2.4 gives an example of KVL. We go around the circuit in the direction of the arrow, which is the direction in which we think current will flow. On passing the battery, the potential increases by 6V. We then lose 4V on passing the 2Ω resistor to give a new potential of 2V. Finally, the potential drops by 2V in the 1Ω resistor. So, the total voltage inside a loop is zero.

$$6V - 4V - 2V = 0$$

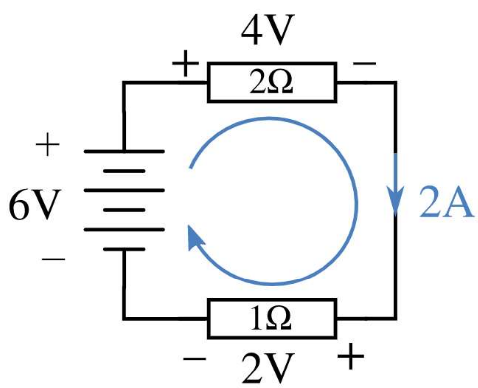
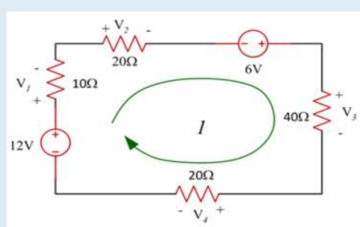


Fig. 2.4

Example 2.1

Find the current *I* and voltage *V* over each resistor.



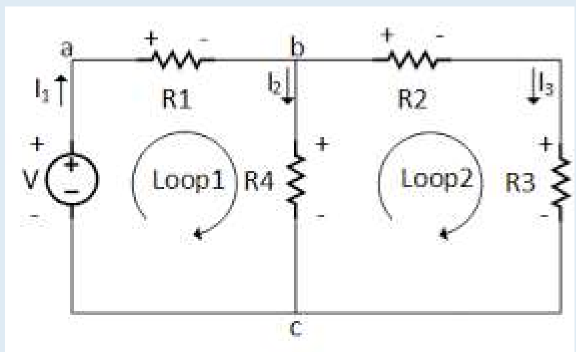
Solution:
Apply KVL:
$12V - V_1 - V_2 + 6V - V_3 - V_4 = 0$
Putting the values of Voltages using Ohm's Law:
12V – (I X 10 Ω) - (I X 20 Ω) + 6V - (I X 40 Ω) - (I X 20 Ω) = 0
I X 90 Ω = 18V
I = 18V / 90 Ω = 0.2 A
Now Applying Ohm's law to find voltages:
V_1 = 0.2 X 10 = 2 V
V_2 = 0.2 X 20 = 4 V
V_3 = 0.2 X 40 = 8V
V_4 = 0.2 X 20 = 4V

Note for teacher:

- Give an example of circuit containing multiple voltage sources. Ask the students to apply KVL in the circuit.

Example 2.2

Find the current *I* and voltage *V* over each resistor, where as
$R_1 = 5\Omega$, $R_2 = 10\Omega$, $R_3 = 5\Omega$, $R_4 = 10\Omega$
And V = 20Volts



Solution:

There are two loops (closed paths) in the circuit, loop 1 with two resistors and a single voltage source, wherein loop 2 there is no voltage source, three resistors only.
The next step is to write equations for each loop. Based on the sign and current name assigned, as shown below.
Loop 1:
$V = I_1 R_1 + I_2 R_4$
$0 = I_3 R_2 + I_3 R_3 - I_2 R_4$
Notice the negative sign in the second equation, it is because of being opposite in direction of loop arrow i.e., for R_2 and R_3 it is (+ -) but for R4 it is (- +).
It should be noted that
$I_1 = I_2 + I_3$
Now, put the value of I_1, which will give you two simultaneous equations with unknown variable I_2 and I3. The value of these currents can be easily computed.
$V = (R_1 + R_4) I_2 + R_1 I_3$
$0 = - R_4 I_2 + (R_3 + R_2) I_3$
Let's put the values of resistors and the voltage source,

$$20 = (5 + 10)I_2 + 5I_3 — -(1)$$

$$0 = -10I_2 + (5 + 10)I_3 — - (2)$$

Now multiply the equation (1) by -3 and add with the equation (2)

$$0 = -10I_2 + (15)I_3 — -(2)$$

$$-60 = -45I_2 - 15I_3 — -(1)$$

Adding the above two equations will give us the equation (3)

$$-60 = -55I_2 + 0 - (3)$$

$$I_2 = -\frac{60}{-55} = 1.09Amp$$

Now putting the current I2 in the equation (2) to get current I3

$$0 = -10(1.09) + 15I_3$$

$$I_3 = 10\frac{(1.09)}{15} = 0.72Amp$$

By applying KCL to the node b, the following equation can be obtained,

$$I_1 = I_2 + I_3$$

Now putting the values of the current will give us the value of I1

$$I_1 = 1.09 + 0.72 = 1.81Amp$$

Now, it is easy to find any parameter of the circuit, just use the ohm's law, as shown below

$$V_3 = I_3R_3 = 0.72(5) = 3.6Volts$$

$$V_2 = I_3R_2 = 0.72(10) = 7.2Volts$$

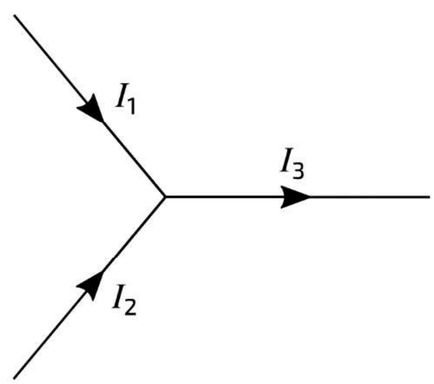
$$V_4 = I_2R_4 = 1.09(10) = 10.9Volts$$

$$V_1 = I_1R_1 = 1.81(5) = 9.05Volts$$

2.3 Kirchhoff's Current Law

Kirchhoff's Current Law (KCL) states that the current flowing towards a node (or a junction) must be equal to the current flowing away from it. This is a consequence of charge conservation.

Current flow in circuits occurs when charge carriers travel around the circuit. Current is defined as the rate at which this charge passes any point in the circuit. A fundamental concept in physics is that charge will always be conserved. In the context of circuits this means that, since current is the rate of flow of charge, the current flowing into a point must be the same as current flowing out of that point.

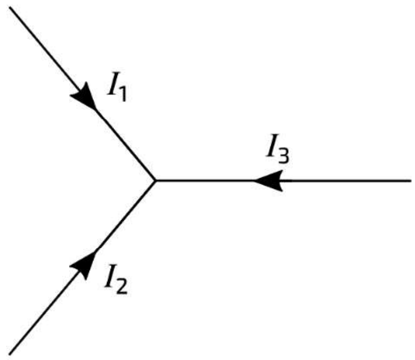


Applying KCL on figure:

$$I_1 + I_2 = I_3$$

It is important to note what is meant by the signs of the current in the diagram. A positive current means that the currents are flowing in the directions indicated on the diagram. Directions are the direction in which any positive charges would flow, that is from the plus of the battery round the circuit to the minus.

The standard way of displaying Kirchhoff's current law is by having all currents either flowing towards or away from the node, as shown in Figure 2:

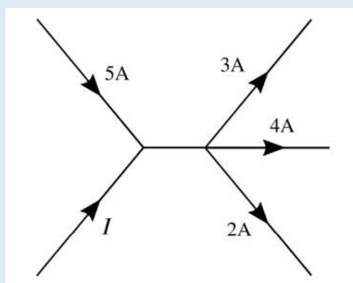


Here, at least one of the currents will have a negative value (in the opposite direction to the arrows on this diagram) and Kirchhoff's current law here would be written as:

$$I_1 + I_2 + I_3 = 0$$

Example 2.3

What is the value of I in the circuit segment shown in Figure?



Solution:

Apply KCL:

$I + 5A = 3A + 4A + 2A$

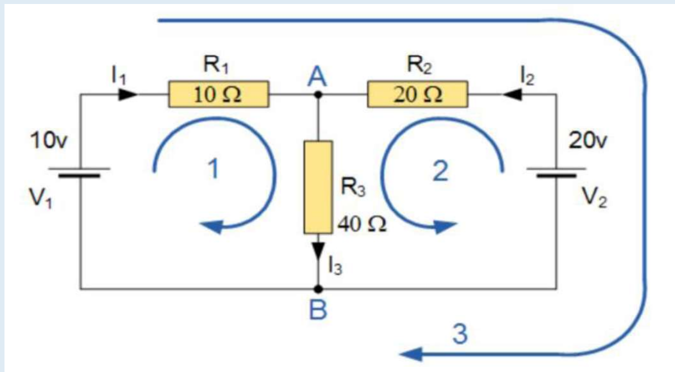
$I = 4A$

Interesting Information

- In 1845 Kirchhoff first announced Kirchhoff's laws, which allow calculation of the currents, voltages, and resistances of electrical networks. Extending the theory of the German physicist Georg Simon Ohm, he generalized the equations describing current flow to the case of electrical conductors in three dimensions. In further studies, he demonstrated that current flows through a conductor at the speed of light.

Example 2.4

Find the current flowing in the 40Ω Resistor, R_3



Solution:

The circuit has 3 branches, 2 nodes (A and B) and 2 independent loops.

Using Kirchhoffs Current Law, KCL the equations are given as:

At node A : $I_1 + I_2 = I_3$

At node B : $I_3 = I_1 + I_2$

Using Kirchhoffs Voltage Law, KVL the equations are given as:

Loop 1 is given as : $10 = R_1 I_1 + R_3 I_3 = 10 I_1 + 40 I_3$

Loop 2 is given as : $20 = R_2 I_2 + R_3 I_3 = 20 I_2 + 40 I_3$

Loop 3 is given as : $10 - 20 = 10 I_1 - 20 I_2$

As I3 is the sum of I1 + I2 we can rewrite the equations as;

Eq. No 1 : $10 = 10 I_1 + 40(I_1 + I_2) = 50 I_1 + 40 I_2$

Eq. No 2 : $20 = 20 I_2 + 40(I_1 + I_2) = 40 I_1 + 60 I_2$

We now have two "Simultaneous Equations" that can be reduced to give us the values of I_1 and I_2

Substitution of I_1 in terms of I_2 gives us the value of I_1 as -0.143 Amps

Substitution of I_2 in terms of I_1 gives us the value of I_2 as +0.429 Amps

As : $I_3 = I_1 + I_2$

The current flowing in resistor R_3 is given as : $-0.143 + 0.429 = 0.286$ Amps

and the voltage across the resistor R_3 is given as : $0.286 \times 40 = 11.44$ volts

The negative sign for I_1 means that the direction of current flow initially chosen was wrong, but never the less still valid. In fact, the 20v battery is charging the 10v battery.

Key Points

- An Electric Circuit is a closed path to allow the flow of electrons.
- A voltage source acts as a source of electrons and hence causes current.
- A voltage is a potential difference between two points of an electric circuit.
- Resistors are commonly used to reduce the current flowing in a certain branch of an electric circuit.
- A capacitor stores electrical energy in its electric field.
- A capacitor blocks the DC but allows AC to pass through it.
- In series connection, there is a single path for the current flow.
- The sum of current in each branch of a parallel circuit is equal to the total current flowing in the circuit.
- Kirchhoff's Voltage Law (KVL) states that the sum of voltages inside a closed loop in an electric circuit is always equal to zero.
- Kirchhoff's Current Law (KCL) states that the current flowing towards a node (or a junction) must be equal to the current flowing away from it.

Exercise

Multiple Choice Questions

1. KVL states that sum of ______ in loop is always zero.
   a. Resistance b. Voltage c. Current d. None of these
2. KCL states that sum ______ flowing at a node is always zero.
   a. Resistance b. Voltage c. Current d. None of these
3. _____ can store electrical charge.
   a. Resistance b. Capacitor c. Inductor d. None of these
4. In series connection following remains the same
   a. Resistance b. Voltage c. Current d. None of these
5. In parallel connection following remains the same
   a. Resistance b. Voltage c. Current d. None of these

Write short answer of the following

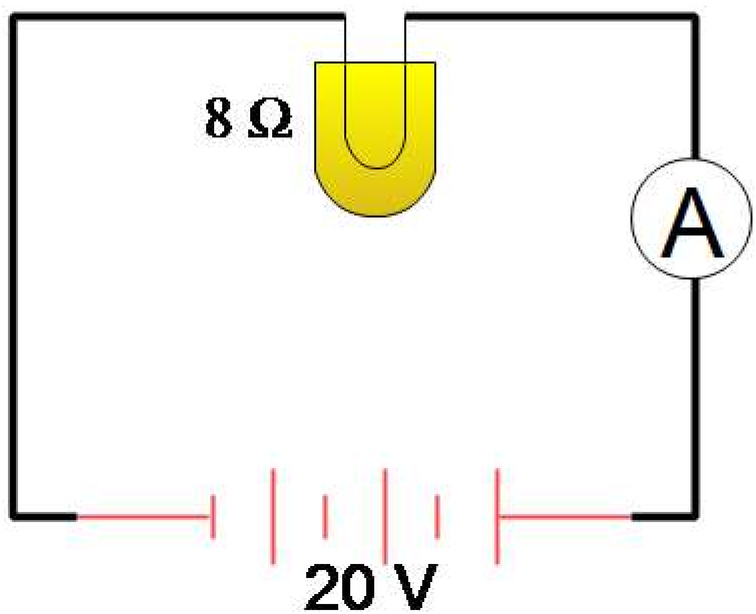
1. What is an electric circuit?
2. What are the components in an electric circuit?
3. Define Voltage, Resistance and Current.
4. What is a capacitor?
5. Differentiate between series and parallel circuits.

Answer the following question in detail.

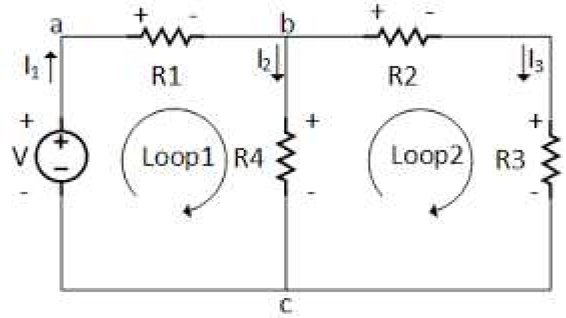
1. Describe KCL.
2. Describe KVL.

Solve the Following

1. If the resistance of an electric hair dryer is 80 Ω and a current of 2.2 A flows through the resistance. Find the voltage between two points.
2. An electronic device has a resistance of 20 ohms and a current of 15 A. What is the voltage across the device?
3. **3.** In the circuit shown below, how much current does the ammeter show?



4. Find the current *I* and voltage *V* over each resistor, where R_1 = 13Ω, R_2 = 15Ω, R_3 = 5.5Ω, R_4 = 11Ω and V = 25Volts.



5. Find the voltages across R_1, R_2 and R_3.